

Voltage Stepdown - Easy Problem

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You are designing a jacuzzi for your cat that requires a 5VDC power supply that draws 3A of continuous current. However, you only have a car battery that can supply only 12VDC to your board.

(Optional Challenge) Your only car battery is also quite unreliable and is not guaranteed to be constantly supplying the nominal 12VDC, though you may assume that it will always be able to supply the necessary current.

Design a circuit that goes in between the car battery and your cat jacuzzi to resolve this issue.